

CURRICULUM VITAE

Daniel Brenner

Phonetician, Software Engineer



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U.S. Citizen

Education

2015	PhD, Department of Linguistics, University of Arizona, Tucson. Phonetics, Cognitive Science.
2011	M.S., Department of Linguistics, University of Arizona, Tucson. Human Language Technology.
2008	M.A., Department of East Asian Studies, University of Arizona, Tucson. Japanese Linguistics.
2003	B.A., Department of East Asian Studies, Oakland University, Rochester, Michigan. East Asian Studies.
2003	Certification, Teaching English as a Second Language, Department of Linguistics, Oakland University, Rochester, Michigan.
2000 – 2002	Exchange student, Nanzan University, Nagoya, Japan. Japanese language and cultural studies.

Employment

- Spring 2026 *Lecturer*, Human Language Technology Program, Department of Linguistics, University of Arizona, Tucson, AZ (remote). Taught on-line “LING 538: Computational Linguistics”, “LING 581: Advanced Computational Linguistics”, and “LING 696G: Advanced Speech Technology”.
- May 2025 – present *Independent Software Engineer*. Designed and built a Flask web application for Gage R&R measurement system analysis in manufacturing. Implemented variance component estimation and statistical tests to assess repeatability and reproducibility across operators, parts, and trials, with reporting aligned to AIAG and Six Sigma standards.
- Spring 2025 *Lecturer*, Human Language Technology Program, Department of Linguistics, University of Arizona, Tucson, AZ (remote). Taught on-line “LING 539: Statistical Natural Language Processing”.

2020 – 2024

Associate AI Engineer, ETS, San Francisco, CA, and remote.

- Architected a generalized scoring engine framework that became the base for 4 language learning applications deployed in the China and India markets.
- Trained and deployed updated speech and text scoring models for annual TOEFL and TOEIC assessment cycles.
- Reduced user-facing latency of speech services by 80% through caching, vectorization, and systematic operation-level latency analysis.
- Built user behavior simulations to optimize infrastructure scaling, enabling 200 concurrent users through language learning progressions.
- Led development of speech feedback features including sentence stress, lexical stress, and speech rhythm analysis.
- Led a team of 2–3 engineers on speech feature development.
- Served as liaison between Infrastructure/Architecture and product development teams, driving prioritization and division of labor across groups.
- Chaired the AI Ethics Working Group, compiling resources and methods for ethical application of AI within the engineering labs.
- Coordinated the AI Ethics Lecture Series, inviting 10 outside experts to present research on topics central to the lab's work.

2017 – 2019

Manager, Living Zen Organics, Detroit, MI. Managed, cooked for, and did accounting for an organic food co-op supporting a Buddhist community.

2015 – 2017

Postdoctoral Research Fellow, Alberta Phonetics Laboratory, University of Alberta, Edmonton, Canada. Speech production and perception research.

2009 – 2010

Research Fellow, Radboud University, Nijmegen, the Netherlands; Sound to Sense project, European Union Marie Curie Research Training Network. Preparation of spontaneous speech corpora for phonetic research, development of automated methods of speech labeling.

2007 – 2010	<i>Instructor</i> (summer and winter sessions), East Asian Studies Department, University of Arizona, Tucson. “TRAD 101: Writing Systems of the World”.
2006 – 2008	<i>Independent Japanese Translator.</i> Various corporate and private translations.
2005	<i>Long-term Substitute Instructor of Japanese and Mandarin,</i> Hutchins Middle School, Detroit, MI, U.S.A.
2004 – 2005	<i>Teaching Coordinator and English Instructor,</i> Guangxi University, Nanning, P.R. China.
2003 – 2004	<i>English Instructor,</i> Guizhou University, Guiyang, P.R. China.
2003	<i>Para-pro,</i> Birmingham Public Schools, Birmingham, Michigan. Japanese interpreting and care for autistic children.

Other Academic Positions

2014 – 2015	<i>Graduate Research Associate,</i> Auditory Cognitive Neuroscience Experience Lab (Headed by Andrew Lotto), University of Arizona, Tucson. Auditory perception research.
2013	<i>Research Fellow,</i> American National Science Foundation East Asia and Pacific Summer Institutes (EAPSI), Beijing, P.R. China. Mandarin tone production and perception research, Chinese Academy of Social Sciences Laboratory of Phonetics and Speech Science.
2011 – 2014	<i>Graduate Research Associate,</i> “Cross-linguistic Language Variation”, Douglass Phonetics Laboratory (Headed by Natasha Warner), University of Arizona, Tucson. Phonetics research.
2010 – 2014	<i>Graduate Research Associate,</i> “Instrumental and Experimental Analysis of Scottish Gaelic Sound Structures” (Headed by Andrew Carnie), University of Arizona, Tucson. Phonetic and psycholinguistic research.
2006 – 2010	<i>Graduate Assistant in Teaching,</i> Linguistics & East Asian Studies Departments, University of Arizona, Tucson.

Dissertation

Daniel Brenner. *The phonetics of Mandarin tones in conversation*. PhD thesis, University of Arizona, 2015.

Publications

Viktor Kharlamov, Daniel Brenner, and Benjamin V Tucker. Individual-level patterns in the random forest classification of fricatives in conversational English. *JASA Express Letters*, 6(2):025205, 2026.

Natasha Warner, Daniel Brenner, Benjamin V Tucker, and Mirjam Ernestus. Non-native listeners' use of information in parsing ambiguous casual speech. *Languages*, 10(1):8, 2025.

Viktor Kharlamov, Daniel Brenner, and Benjamin V Tucker. Examining the effect of high-frequency information on the classification of conversationally produced English fricatives. *The Journal of the Acoustical Society of America*, 154(3):1896–1902, 2023.

Viktor Kharlamov, Daniel Brenner, and Benjamin V Tucker. Temporal and spectral characteristics of conversational versus read fricatives in American English. *The Journal of the Acoustical Society of America*, 152(4):2073–2081, 2022.

Natasha Warner, Dan Brenner, Benjamin V Tucker, and Mirjam Ernestus. Native listeners' use of information in parsing ambiguous casual speech. *Brain Sciences*, 12(7):930, 2022.

Benjamin V Tucker, Daniel Brenner, D Kyle Danielson, Matthew C Kelley, Filip Nenadić, and Michelle Sims. The massive auditory lexical decision (MALD) database. *Behavior research methods*, pages 1–18, 2018.

Adam Ussishkin, Natasha Warner, Ian Clayton, Daniel Brenner, Andrew Carnie, Michael Hammond, and Muriel Fisher. Lexical representation and processing of word-initial morphological alternations: Scottish gaelic mutation. *Laboratory Phonology*, (22), 2017.

Natasha Warner, Daniel Brenner, Jessamyn Schertz, Andrew Carnie, Muriel Fisher, and Michael Hammond. The aerodynamic puzzle of nasalized fricatives: Aerodynamic and perceptual evidence from Scottish Gaelic. *Laboratory Phonology*, 6(2):197–241, 2015.

Daniel Brenner, Andrea Davis, Natasha Warner, Andrew Carnie, Muriel Fisher, Jessamyn Schertz, Michael Hammond, and Diana Archangeli. Can you say [ṽ] or [x̃]? Aerodynamics of nasalized fricatives in Scottish Gaelic. *The Journal of the Acoustical Society of America*, 130(4):2550, 2011. Poster presentation.

Proceedings

Yoichi Mukai, Daniel Brenner, and Benjamin V Tucker. Durational variability of spontaneous and read speech: Comparison between english and japanese. In *Proceedings of Meetings on Acoustics*, volume 56, pages 1–12. Acoustical Society of America, 2025.

Rahul R. Divekar, Haley Lepp, Pravin Chopade, Aaron Albin, Daniel Brenner, and Vikram Ramnarayanan. Conversational agents in language education: Where they fit and their research challenges. In *HCI International 2021 - Late Breaking Papers: Multimodality, eXtended Reality, and Artificial Intelligence*, pages 272–279, Washington DC, USA, 2021. Springer.

Catherine Ford, Filip Nenadić, Daniel Brenner, and Benjamin V Tucker. Shorter phone duration facilitates isolated spoken word recognition. *Proceedings of The 11th International Conference on the Mental Lexicon*, 2019.

Ayushi Mrigen, Daniel Brenner, and Benjamin V Tucker. What do forced alignment likelihood scores tell us about the aligned speech? *Canadian Acoustics*, 44(3), 2016.

Helena Spilková, Daniel Brenner, Anton Öttl, Pavel Vondříčka, Wim Van Dommelen, and Mirjam Ernestus. The Kachna L1/L2 picture replication corpus. In *Seventh conference on International Language Resources and Evaluation (LREC)*, pages 2432–2436. European Language Resources Association (ELRA), 2010.

Software

auditoryMaskedPrimingStimGenerator, Praat script performing acoustic manipulations to construct auditory masked priming stimuli (Kouider & Dupoux 2005).

EenTweeDrie, a wrapper and framework for forced alignment of speech corpora using Young et al.'s (2009) Hidden Markov Model Toolkit.

Other assorted scripts at:

<https://www.github.com/dbqpdb>

Oral Presentations

- Ki Woong Moon and Daniel Brenner. Enhancing HuBERT for spontaneous speech with acoustic feature fusion. *The Journal of the Acoustical Society of America*, 158(4_Supplement):A259, 2025. Oral presentation.
- Daniel Brenner. Exploring individual differences in auditory lexical decision. In *Auditory Cognitive Neuroscience Society meeting*, volume 11, Gainesville, Florida, USA, January 2017.
- Benjamin V Tucker and Daniel Brenner. Massive auditory lexical decision: Going big in the auditory domain. In *International Conference on the Mental Lexicon*, volume 10, Ottawa, ON, Canada, Oct 20 2016.
- Viktor Kharlamov, Daniel Brenner, and Benjamin V Tucker. The acoustics of conversational fricatives in mid-Western American English. In *LabPhon: "Reduction" special section*, Cornell University, Ithaca, NY, USA, July 13 2016.
- Benjamin V Tucker, Daniel Brenner, and Viktor Kharlamov. The acoustics of conversational fricatives in mid-Western American English. In *North West Phonetics & Phonology Conference (NoWPhon)*, Eugene, OR, U.S.A., 2016.
- Daniel Brenner. Getting friendly with the giants: The phonetics of Mandarin conversational tones. In *University of Alberta Linguistics Colloquium*, Edmonton, Alberta, Canada, March 11 2016.
- Alejandra Baltazar Molina, Yamile Andrea Díaz Gutiérrez, and Daniel Brenner. Una perspectiva estudiantil de las investigaciones fonéticas realizadas por estudiantes de la universidad de arizona [a students' perspective of phonetic research at the University of Arizona]. In *Autonomous University of Querétaro, Querétaro, Mexico*, Mar 21 2014.
- Daniel Brenner. Tone production & perception in conversation. In *China-U.S. Young Scientists Forum*, Beijing, PR China, Aug 2 2013.
- Natasha Warner, Daniel Brenner, and Andrew Carnie. Nasal fricatives vs. approximants in Scottish Gaelic. In *Celtic Linguistics Conference*, volume 7, Rennes, France, June 2011.
- Daniel Brenner, Natasha Warner, Andrew Carnie, Muriel Fisher, Diana Archangeli, Michael Hammond, Micaya Clymer, Andréa Davis, Colin Gorrie, Julia Fisher, Lionel Mathieu, Jessamyn Schertz, and Jae-Hyun Sung. Fitting the phonetically impossible into phonology: Nasalized fricatives in Scottish Gaelic. In *Department of Linguistics Spring Datablitz*, University of Arizona, Tucson, AZ, USA, Feb 11 2011.
- Natasha Warner, Andrew Carnie, Daniel Brenner, Micaya Clymer, Lionel Mathieu, Jae-Hyun Sung, Jessamyn Schertz, Diana Archangeli, Michael Hammond, Muriel Fisher, and Colin Gorrie. Nasalization and frication in Scottish Gaelic. In *International Congress of Celtic Studies*, volume 14, Maynooth, Ireland, August 2011.
- Andrea Davis, Michael Hammond, Andrew Carnie, Muriel Fisher, Diana Archangeli, Natasha Warner, Daniel Brenner, Micaya Clymer, Julia Fisher, Colin Gorrie, Lionel Matthieu, Jessamyn Schertz, and Jae-Hyun Sung. Syllable count of epenthetic words in Scottish Gaelic. In *Department of Linguistics Spring Datablitz*, University of Arizona, Tucson, AZ, USA, Feb 11 2011.

Andrew Carnie, Diana Archangeli, Muriel Fisher, Michael Hammond, Natasha Warner, Daniel Brenner, Micaya Vance Clymer, Andrea Davis, Colin Gorrie, Lionel Mathieu, Jessamyn Schertz, and Jae-Hyun Sung. Scottish Gaelic experimental phonology: A preliminary report. invited plenary talk. In *Celtic Linguistics Conference*, volume 6, University College, Dublin, Ireland, Sep 12 2010. Plenary talk.

Poster Presentations

Viktor Kharlamov, Daniel Brenner, and Benjamin V Tucker. Classification patterns in conversational English fricatives: Between- and within-speaker analyses. *The Journal of the Acoustical Society of America*, 155(3_Supplement):A313–A314, 2024. Poster presentation.

Viktor Kharlamov, Daniel Brenner, and Benjamin V Tucker. Extended high frequencies for fricative classification in conversational speech. *The Journal of the Acoustical Society of America*, 153(3_Supplement):A290, 2023. Poster presentation.

Catherine Ford, Filip Nenadić, Daniel Brenner, and Benjamin V Tucker. Effects of the variation of phoneme duration on word processing. *The Journal of the Acoustical Society of America*, 141(5_Supplement):4038, 2017. Poster presentation.

Benjamin V Tucker and Daniel Brenner. Exploring the acoustic characteristics of individual variation. *The Journal of the Acoustical Society of America*, 141(5_Supplement):3579, 2017. Poster presentation.

Jae-Hyun Sung and Daniel Brenner. Three-dimensional reduction for three-way contrast: Conversational stops in Korean. *The Journal of the Acoustical Society of America*, 140(4_Supplement):3111, 2016. Poster presentation.

Benjamin V Tucker and Daniel Brenner. MALD: Massive Auditory Lexical Decision. In *LabPhon*, volume 15, Cornell University, Ithaca, NY, USA, July 15 2016.

Daniel Brenner and Benjamin V Tucker. Acoustic reduction, context, and inter-stimulus interval in cross-modal priming. In *171st Meeting of the Acoustical Society of America*, Salt Lake City, UT, U.S.A., 2016.

Natasha Warner, Ian D Clayton, Andrew Carnie, Muriel Fisher, Daniel Brenner, and Mike Hammond. The effect of Gaelic initial consonant mutation on spoken word recognition. In *Celtic Linguistics Conference*, volume 8, Edinburgh, UK, June 6 2014.

Daniel Brenner. Acoustic separability of Mandarin tonal categories in careful and conversational speech. In *Acoustical Society of America meeting*, Providence, RI, USA, May 9 2014.

Benjamin V Tucker, Daniel Brenner, and Michelle Sims. Medial stop reduction and word recognition “in the wild”. In *International Conference on the Mental Lexicon*, volume 9, Niagara-on-the-Lake, ON, Canada, Sep - Oct 2014.

Jae-Hyun Sung, Diana Archangeli, Ian Clayton, Daniel Brenner, Samuel Johnston, Michael Hammond, and Andrew Carnie. The articulation of lexical palatalization in Scottish Gaelic. In *Acoustical Society of America meeting*, Providence, RI, USA, May 9 2014.

- Natasha L. Warner, Miguel Simonet, Benjamin V. Tucker, Daniel Brenner, Maureen Hoffmann, Alejandra Baltazar, Andrea Morales, and Yamile Diaz. Spontaneous speech variability across languages: Labial and velar stops. In *Acoustical Society of America meeting*, Providence, RI, USA, May 8 2014.
- Daniel Brenner. The acoustics of Mandarin tones in careful and conversational speech. In *Acoustical Society of America meeting*, San Francisco, CA, USA, Dec 6 2013.
- Miguel Simonet, Natasha L. Warner, Daniel Brenner, Maureen Hoffman, Andrea Morales, and Alejandra Baltazar Molina. Range of variability in native and non-native spontaneous speech intervocalic stops. In *Acoustical Society of America meeting*, San Francisco, CA, USA, Dec 6 2013.
- Natasha L Warner, Ian Clayton, Andrew Carnie, Muriel Fisher, Daniel Brenner, Michael Hammond, Diana Archangeli, and Adam Ussishkin. Perception of Scottish Gaelic alternating (leniting) consonants. In *Acoustical Society of America meeting*, San Francisco, CA, USA, Dec 3 2013.
- Ian D Clayton, Diana Archangeli, Dan Brenner, Andrew Carnie, Mike Hammond, Samuel Johnston, and Jae-Hyun Sung. The articulation of Scottish Gaelic plain and palatalized consonants. In *UltraFest*, volume 4, Edinburgh, UK, Nov 7 2013.
- Samuel Johnston, Daniel Brenner, Rolando Coto, and Diana Archangeli. Probe skewing effects & kinect head-tracking in tongue contour imaging. In *UltraFest*, volume 4, Edinburgh, UK, Nov 6 2013.
- Katherine M Carbonell, Daniel Brenner, and Andrew J Lotto. Discriminating languages with general measures of temporal regularity and spectral variance. In *Acoustical Society of America meeting*, Kansas City, MO, USA, Oct 24 2012.
- Natasha L Warner, Daniel Brenner, Benjamin V Tucker, Jae-Hyun Sung, Mirjam Ernestus, Miquel Simonet, and Ana Gonzalez. Processing reduced speech across languages and dialects. In *Acoustical Society of America meeting*, Kansas City, MO, USA, October 2012.
- Daniel Brenner. Timing differences in read speech and spontaneous conversation: English, Japanese, Korean, and Mandarin. In *Acoustical Society of America meeting*, San Diego, CA, USA, November 2011.
- Daniel Brenner, Natasha Warner, Mirjam Ernestus, Benjamin V Tucker, and Jaycie Martin. Parsing the ambiguity of casual speech: ‘he was like’ or ‘he’s like’? In *Acoustical Society of America meeting*, Seattle, WA, USA, May 2011.
- Daniel Brenner, Natasha Warner, Mirjam Ernestus, and Benjamin V. Tucker. Integration of bottom-up and high-level cues in parsing spontaneous conversational speech. In *Arizona State University / University of Arizona Cognitive Science Conclave*, Tempe, AZ, USA, November 2010.
- Natasha Warner, Daniel Brenner, Anna Woods, Benjamin V. Tucker, and Mirjam Ernestus. ‘Were we’ or ‘are we’?: Perception of reduced function words in spontaneous conversations. In *Acoustical Society of America meeting*, Portland, OR, USA, May 2009.

Teaching

At the University of Alberta:

Statistics practice group, Alberta Phonetics Lab. Summer, 2016.

Using Speech in Psycholinguistics, at Spring Training in Experimental Psycholinguistics (STEP), 16–21 May, 2016. Co-taught with Benjamin V. Tucker.

At the University of Arizona:

Linguistics 696G: Advanced Speech Technology, Online course.

Linguistics 538: Computational Linguistics, Online course.

Linguistics 539: Statistical Natural Language Processing, Online course.

Linguistics 101: Language (formerly “Individuals and Societies 101: Mind, Self, & Language”), Online course.

Traditions and Cultures 101: Writing Systems of the World.

Traditions and Cultures 101: Languages and Cultures of East Asia (taught discussion sections only).

Japanese 101: Introductory Japanese.

Linguistics 123: Introduction to Mathematical Approaches to Language

At Hutchins Middle School:

Japanese.

Mandarin.

At Guangxi University:

English Conversation.

English Composition.

Current Events.

At Guizhou University:

English Conversation.

Reviewer for

- Journal of the Acoustical Society of America
- Laboratory Phonology
- Language & Speech
- Speech Prosody
- Mental Lexicon
- Arizona Linguistics Circle

Languages

- English (native)
- Japanese
- Mandarin

Other Service and Research Experience

2023 – 2024	<i>Coordinator</i> , AI Ethics Lecture Series, ETS AI Labs, San Francisco, CA.
2022 – 2023	<i>Chair</i> , AI Ethics Best Practices Working Group, ETS AI Labs, San Francisco, CA.
2015	<i>Curriculum Committee Member</i> , planning and materials development for the new course, LING 123 “Introduction to Mathematical Approaches to Language”, Department of Linguistics, University of Arizona, Tucson.
2012 – 2013	<i>Awards Committee Member</i> , Department of Linguistics, University of Arizona, Tucson.
2012	<i>Organizing committee chair</i> , Arizona Linguistics Circle Conference 6, University of Arizona, Tucson.
2012	<i>Editor</i> , Coyote Papers: Working Papers in Linguistics, University of Arizona Linguistics Circle, Tucson.
2011	<i>Organizing committee member</i> , Arizona Linguistics Circle Conference 5, University of Arizona, Tucson.
2011	<i>Organizing committee member</i> , West Coast Conference on Formal Linguistics 29, University of Arizona, Tucson.
2011	<i>Editor</i> , Coyote Papers: Working Papers in Linguistics, University of Arizona Linguistics Circle, Tucson.
2010	<i>Organizing committee member</i> , Arizona Linguistics Circle Conference 4, University of Arizona, Tucson.
2010	<i>Treasurer</i> , Arizona Linguistics Circle, University of Arizona, Tucson.

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| 2007 – 2008 | <i>Laboratory manager</i> , Linguistics laboratory, Department of East Asian Studies, University of Arizona, Tucson. |
| 2007 – 2008 | <i>Departmental graduate representative</i> , Department of East Asian Studies, University of Arizona, Tucson. |
| 2002 – 2003 | <i>President</i> , China Club student organization, Oakland University, Rochester, Michigan. |
| 1999 – 2000 | <i>President</i> , Japan Club student organization, Oakland University, Rochester, Michigan. |